Predictive Parsing (2)

Sam Ogden

DRAFT

This slide deck is still under active development. Expect changes before it is finalized.

Learning outcomes

- After this lecture, you should be able to:
- derive parse trees from a BNF grammar
- define what it means for a grammar to be “ambiguous”
- use two additional rules for transforming a grammar

Parse trees

```
expr ::= num | var | expr+expr
```

Parse `1 + x`

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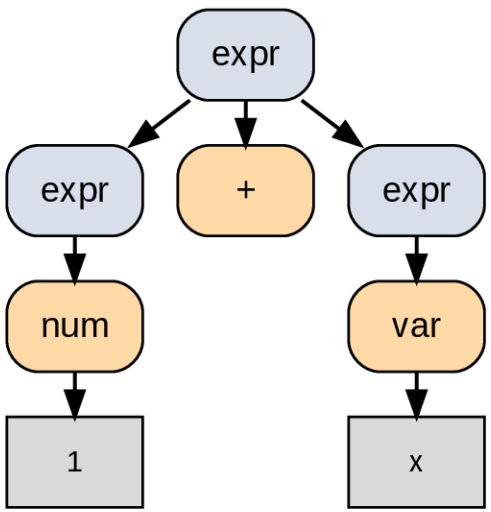
Parse `1 + x`



Parse trees

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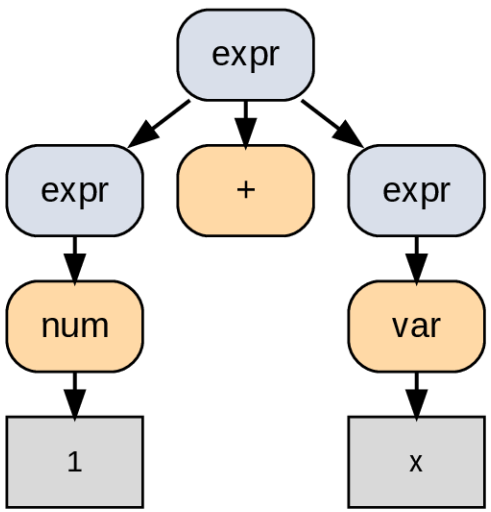
expr ::= num | var | expr+expr
```

Parse `1 + x`



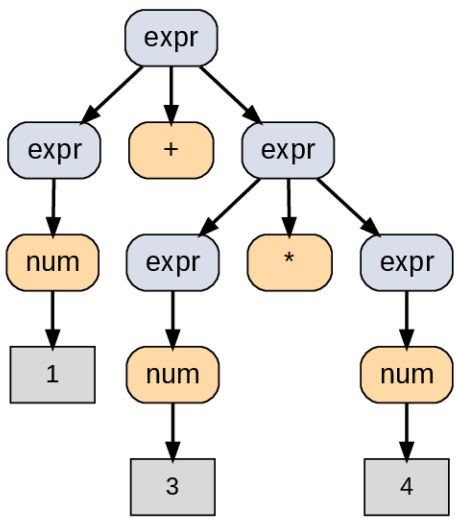
Rules:

1. root is labeled with the start symbol
2. the symbols in one of its productions become child nodes
3. continue until all leaf nodes are terminal symbols

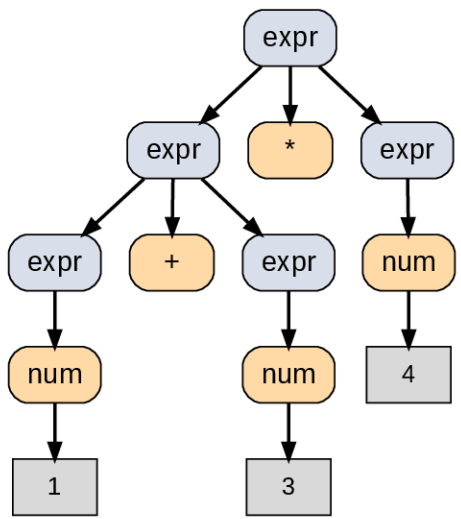
Ambiguous grammar

```
expr ::= num | expr+expr | expr*expr
```

Create a parse tree for `1 + 3 * 4`:



`1 + (3 * 4)`



`(1 + 3) * 4`

Ambiguous grammar

- A grammar is **ambiguous** if, for some string in its language, there is more than one parse tree.
- Different parse trees usually suggest different meanings.
- So: we usually want unambiguous grammars.

Review: predictive parsing

```
stmt ::= print(expr) | id=expr
expr ::= id | num
```

- We need to be able to pick a production of a non-terminal based on the next token.

```
1  void stmt() {
2    switch (token) {
3      case PRINT:
4        match(PRINT); match("("); expr(); match(")");
5        break;
6      case ID:
7        match(ID); match("="); expr();
8        break;
9      default:
10       error("syntax error");
11   }
12 }
```

Transforming a BNF grammar

- In the last lecture we learned how to deal with:
- left recursion
- empty productions

Two more ways to transform BNF:

- left factoring
- expanding a non-terminal

It's important to get comfortable with modifying grammars.

Left factoring

```
A ::= aB | aC | bD
```

Left factoring

```
A ::= aB | aC | bD
```

How about this replacement? Is it the same?

```
A ::= aE | bD
E ::= B | C
```

Exercise

Suppose an expression can be a variable or a function call.

- Examples: `tot` (variable), `max(y)` (function call)

BNF:

```
expr ::= var(expr) | var
```

Exercise: Apply left factoring.

Exercise

Suppose an expression can be a variable or a function call.

- Examples: `tot` (variable), `max(y)` (function call)

BNF:

```
expr ::= var(expr) | var
```

Exercise: Apply left factoring.

Solution

```
expr ::= varexpr1
expr1 ::= (expr) | ""
```

Expanding a non-terminal

```
A ::= aB | E
E ::= cC | dD
```

Any problems with a predictive parser here?

Expanding a non-terminal

```
A ::= aB | E
E ::= cC | dD
```

Any problems with a predictive parser here?

Solution

```
A ::= aB | cC | dD
```

Rewriting BNFs

- For predictive parsers, we want that, for every non-terminal in the grammar:
- there's only one production for it, or
- each production starts with a terminal token (maybe "" on its own) and each of those tokens are different

Exercise

```
prog ::= stmt | stmt;prog
stmt ::= ID=expr | ID(expr)
```

- There are two problems. What are they?
- Apply left factoring.

Solution

```
prog ::= stmtprog1
prog1 ::= ;prog | ""
stmt ::= IDstmt1
stmt1 ::= =expr | (expr)
```

Exercise

Expand a non-terminal (inline `stmt` into `prog`):

```
prog ::= stmtprog1
prog1 ::= ;prog | ""
stmt ::= IDstmt1
stmt1 ::= =expr | (expr)
```

Solution

```
prog ::= IDstmt1prog1
prog1 ::= ;prog | ""
stmt1 ::= =expr | (expr)
```

Note: there is no longer a `stmt` non-terminal.

Fixed grammar

When expanding non-terminals, verify you didn't change the language.

Example check:

- Original `prog ::= stmt | stmt ; prog` does **not** derive `ID = expr ;` (no empty statement at the end).
- So the transformed grammar should not allow a trailing `;` either.

```
prog ::= IDstmt1prog1
prog1 ::= ;prog | ""
stmt1 ::= =expr | (expr)
```

```
Derive ID = expr ;

grammer 1:
prog ::= stmt | stmt ;  prog
stmt ::= ID = expr | ID ( expr )

prog -> stmt -> ID = expr -> FALSE
prog -> stmt ; prog -> ID = exper ; prog -> FALSE
Can't produce a ; at the end. No empty statement.


If we change      stmt ; prog      to      stmt ; prog1
Then we make prog1 ::= "" | stmt
I belive they will be the same then. Deriving the soulution below.

NEW SOLUTION WITH THE ADDITION OF RULES
prog -> stmt ; prog1 -> ID = expr ; prog1 -> ID = expr ; = TRUE

grammer 2:
prog ::= ID stmt1 ; prog1
prog1 ::= "" | ID stmt
stmt :: = = expr | ( expr )

prog -> ID stmt1 ; prog1 -> ID = expr ; prog1 -> ID = expr ; -> TRUE


If we get rid of "" in prog1 for grammar 2 then it would be the same.
I have derived the new test below

NEW SOLUTION WITH REDUCTION OF RULES
prog -> ID stmt1 ; prog1 -> ID = expr ; prog1 -> FALSE
```

Fixed Grammar by an old TA

Summary

- A parse tree shows how a string can be derived from a BNF grammar.
- To use predictive parsing, our BNF syntax must be in a certain form. Tools to get BNF in that form:
- eliminate left recursion
- left factoring
- expanding non-terminals